Advanced Theory — Module 05: Action — Study Questions

1. What is the renormalization group? How does it relate descriptions at different scales?
2. Explain the block transformation. How does it reduce degrees of freedom?
3. What is an RG fixed point? What physical phenomena does it correspond to?
4. How does Active Inference claim to be scale-free? What mathematical structure is preserved across scales?
5. Identify the Markov blanket at four different organizational scales (molecular, cellular, organismal, social).
6. How are Markov blankets nested? What does nesting mean mathematically?
7. How does RG coarse-graining correspond to Bayesian Model Reduction?
8. What are sufficient statistics in the RG context? How do they relate to effective generative models?
9. What is the evidence that the brain operates near criticality? (neural avalanches, 1/f noise, etc.)
10. Why would operating at a critical point be computationally advantageous for inference?
11. How might criticality relate to optimal precision allocation?
12. How does the immune system perform "tissue-level inference"?
13. How do cultural norms function as shared generative models at the social scale?
14. What is the mathematical relationship between RG flow and hierarchical Bayesian inference?
15. How does the concept of universality in RG relate to Active Inference's scope claims?
16. What are the limitations of the Markov blanket interpretation at the social scale?
17. How does symmetry breaking in RG relate to the emergence of new organizational structures?
18. What role does the coupling constant play in RG? How does its analogue function in Active Inference?
19. How does the Wilson-Fisher RG relate to continuous phase transitions? What is the neural analogue?
20. Critically evaluate: Is the application of RG to Active Inference rigorous, or is it metaphorical?